CFA® 2026 Level I

Complete Study Guide

Fixed Income & Derivatives

This document is a condensed study guide. Please refer to the official CFA Institute curriculum for complete details.

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Part I

Fixed Income

1 Fixed-Income Instrument Features

1.1 Features of Fixed-Income Securities

LOS 47.a: Describe the features of a fixed-income security.

Why This Matters for the CFA Exam

Fixed-income securities are the largest segment of global financial markets. Understanding their features is essential because:

- Governments and corporations raise trillions of dollars through bond issuance
- Pension funds and insurance companies primarily invest in fixed income to match long-term liabilities
- Bond prices and yields are fundamental to valuation across all asset classes
- Interest rate movements directly impact bond portfolios, making risk management critical

A fixed-income security is an investment that provides a return in the form of fixed periodic interest payments (coupons) and the eventual return of principal at maturity.

Major Types of Fixed-Income Instruments:

- **Loans:** Private, nontradable agreements between a borrower and a lender (e.g., bank loans, mortgages)
- **Bonds:** Standardized, tradable securities representing a debt investment that can be bought and sold in secondary markets

Core Features of Fixed-Income Securities:

Feature	Description and Significance

Issuer	The entity borrowing money. Major issuers include sovereign governments, corporations, local governments, supranational entities (IMF, World Bank), quasi-government entities (Fannie Mae, Ginnie Mae), and special purpose entities (SPEs) for asset-backed securities.
Maturity	The date when the final cash flow is due. Bonds with original maturity ≤ 1 year are **money market securities**; > 1 year are **capital market securities**. **Perpetual bonds** have no maturity date.
Par Value	The principal amount repaid at maturity, typically \$1,000 (corporate) or \$100 (government). Also called face value or principal value.
Coupon Rate	The annual interest rate stated as a percentage of par value. A 5% coupon on a \$1,000 bond pays \$50 annually.
Coupon Frequency	How often interest is paid: annual (Europe), semiannual (U.S. corporates), quarterly, or monthly. Higher frequency means more compounding.
Floating-Rate Notes (FRNs)	Pay interest based on a variable market reference rate (MRR) such as SOFR plus a fixed quoted margin. The coupon resets periodically, reducing interest rate risk.
Zero-Coupon Bonds	Pay no interest during the bond's life. Sold at a discount to par; return is the difference between purchase price and par at maturity. Highest duration and interest rate sensitivity.
Seniority	The priority of claims in bankruptcy. Debt ranks above equity. **Senior debt** has priority over **subordinated (junior) debt**. Higher seniority $\rightarrow$ lower credit risk $\rightarrow$ lower yield.
Contingency Provisions	Embedded options that give the issuer or bondholder rights to take certain actions (call, put, convert). These affect bond pricing and risk.

The inverse relationship between bond prices and yields is fundamental: when market yields rise, existing bond prices fall, and vice versa. This

occurs because the fixed coupon payments become less (or more) attractive compared to newly issued bonds.

On the exam, remember that when interest rates rise, bond prices fall—and longer-maturity bonds fall more than shorter-maturity bonds. This is called interest rate risk or duration risk.

1.2 Bond Indenture and Covenants

LOS 47.b: Describe the contents of a bond indenture and contrast affirmative and negative covenants.

The Legal Framework That Protects Bondholders

A **bond indenture** (also called trust deed) is the legal contract between the bond issuer and bondholders that specifies all terms, conditions, rights, and obligations of both parties. A trustee enforces the indenture on behalf of bondholders.

Sources of Repayment by Issuer Type:

- **Sovereign bonds:** Tax revenues and the ability to create new currency (monetize debt)
- **Municipal/local government bonds:** Property taxes, sales taxes, or user fees from specific projects
- **Secured corporate bonds:** Operating cash flows PLUS specific assets pledged as collateral
- **Unsecured corporate bonds:** Operating cash flows only—a general claim against the company's assets
- **Asset-backed securities (ABS):** Cash flows from a pool of financial assets held by a bankruptcy-remote SPE

Covenants: Contractual Protections

Affirmative Covenants (Actions the Issuer MUST Take):

- Timely payment of interest and principal
- Provision of audited financial statements to bondholders

- Maintenance of adequate insurance on collateral
- Compliance with all laws and regulations
- **Cross-default provision:** Default on any debt obligation automatically triggers default on this bond
- **Pari passu clause:** This bond ranks equally with all other senior unsecured debt

Negative Covenants (Actions the Issuer MUST NOT Take):

- **Negative pledge clause:** Cannot pledge the same assets as collateral for other debt without equally securing this bond
- Restrictions on issuing additional debt, especially debt senior to existing bonds
- Limitations on dividend payments and share repurchases (preserves cash for debt service)
- Restrictions on mergers, acquisitions, or changes in control
- Limitations on investments in unrelated businesses
- **Incurrence test:** Certain actions only permitted if specific financial ratios are satisfied

Covenants protect bondholders by restricting actions that could increase default risk. High-yield bonds typically have more (and stricter) covenants than investment-grade bonds.

2 Fixed-Income Cash Flows and Types

2.1 Cash Flow Structures

LOS 48.a: Describe common cash flow structures of fixed-income instruments and contrast cash flow contingency provisions.

A **bullet structure** is the simplest bond structure where the entire principal is repaid in a single payment at maturity. All periodic payments are pure interest.

Example: 5-Year, 5% Annual Coupon Bullet Bond ($1,000 Par)

Year	**1**	**2**	**3**	**4**	**5**
Payment	$50	$50	$50	$50	$1,050
Principal Remaining	$1,000	$1,000	$1,000	$1,000	$0

An **amortizing loan** structure includes both interest and principal repayment in each period.

Fully Amortizing: Principal fully paid off by final payment. *Example: N=5, I/Y=5, PV=1000, FV=0, CPT PMT = 230.97*

Partially Amortizing (Balloon Payment): Some principal repaid; final balloon payment retires remaining balance. *Example: N=5, I/Y=5, PV=1000, FV=200, CPT PMT = 194.78*

A **sinking fund** requires the issuer to retire a portion of the bond issue periodically before maturity.

- **Advantage:** Reduces credit risk
- **Disadvantage:** Creates reinvestment risk when interest rates fall

Coupon Structures:

Coupon Type	**Description**
Fixed-Rate	Coupon rate stays constant throughout the bond's life.
Floating-Rate Note (FRN)	Coupon = MRR + Quoted Margin. Resets periodically.
Step-Up Coupon	Coupon rate increases according to a predetermined schedule.
Credit-Linked Coupon	Coupon changes when the issuer's credit rating changes.
Payment-in-Kind (PIK)	Coupon paid by issuing more bonds (increasing principal).
Green Bonds	Coupon increases if environmental targets are not met.

Inflation-Linked	Cash flows adjusted for inflation (TIPS in U.S.).
Zero-Coupon	No periodic coupon; sold at discount; return is price appreciation.
Deferred Coupon	Regular payments begin after a specified period.

Embedded Options:

Type	Beneficiary	Description
Callable Bond	Issuer	Right to redeem before maturity at call price. Value = Straight bond - Call option.
Putable Bond	Bondholder	Right to sell back to issuer at put price. Value = Straight bond + Put option.
Convertible Bond	Bondholder	Right to exchange for specified number of common shares.
Warrants	Bondholder	Right to buy common shares at fixed price; can be detached and traded.
CoCo Bond (Contingent Convertible)	Automatic	Converts automatically to equity if trigger event occurs (bank capital ratio falls).

Callable bond value = Straight bond value - Call option value. Putable bond value = Straight bond value + Put option value. Convertible bond value = Straight bond value + Conversion option value.

2.2 Legal, Regulatory, and Tax Considerations

LOS 48.b: Describe how legal, regulatory, and tax considerations affect issuance and trading.

Bond Types by Market:

Type	Description
Domestic Bonds	Issuer domiciled in same country as trading market.
Foreign Bonds	Issuer from different country; traded in local market (Yankee, Samurai, Bulldog).

Eurobonds	Issued outside jurisdiction of any single country; less regulated.
Global Bonds	Trade in Eurobond market and at least one domestic market.
Sukuk Bonds	Sharia-compliant; periodic payments structured as rent, not interest.

Taxation of Bond Income:

- Interest income typically taxed as ordinary income
- Municipal bonds (U.S.): Often tax-exempt at federal level
- Capital gains from bond sales taxed at capital gains rate (often lower)
- Original Issue Discount (OID) bonds: Discount treated as interest income, taxed annually even without cash receipt
- Premium bonds: Some jurisdictions allow premium amortization to reduce taxable interest

3 Fixed-Income Issuance and Trading

3.1 Fixed-Income Market Segments

LOS 49.a: Describe fixed-income market segments and their issuer and investor participants.

Market Classifications:

1. **Type of issuer:** Governments, corporates, special purpose entities (ABS)
2. **Credit quality:** Investment-grade (BBB-/Baa3+) vs High-yield (BB+/Ba1-)
3. **Original maturity:** Short-term (¡1 year), intermediate (1-10 years), long-term (¿10 years)

Investor Positioning:

Investor Type	Typical Positioning
Pension funds & Insurance	Long-term, investment-grade securities
Corporations	Short-term investment-grade (commercial paper, repos)
Central banks	Intermediate-term Treasury notes (monetary policy)
Bond funds & ETFs	Intermediate investment-grade (excluding Treasuries)
Asset managers (higher returns)	High-yield intermediate securities, distressed debt
Banks	Treasuries across maturity spectrum

3.2 Fixed-Income Indexes

LOS 49.b: Describe types of fixed-income indexes.

Differences from Equity Indexes:

1. More constituents (issuers have many bonds vs few equity classes)
2. Higher turnover (bonds mature and are replaced frequently)
3. Larger weights in sovereign bonds (governments are largest issuers)

Index Types:

- **Aggregate indexes:** Broad selection across sectors (e.g., Bloomberg Barclays Aggregate)
- **Narrow indexes:** Focus on geography, credit quality, sector, or maturity
- **ESG indexes:** Incorporate environmental, social, governance factors

3.3 Primary and Secondary Markets

LOS 49.c: Compare primary and secondary fixed-income markets to equity markets.

Primary Markets:

- Newly issued bonds sold to investors
- **Public offering:** Registered with securities regulators
- **Private placement:** Sold only to selected investors
- **Underwritten offering:** Price guaranteed by financial intermediaries
- **Best-efforts offering:** No price guarantee; commission-based
- **Shelf registration:** Registered in aggregate; issued over time as needed
- **Auctions:** Used for government bonds

Secondary Markets:

- Trading of previously issued bonds
- Mostly over-the-counter (OTC) dealer market
- Dealers post bid (buy) and ask (sell) prices; spread = dealer's profit
- **On-the-run bonds:** Most recent issue; most liquid; narrowest spreads
- **Distressed debt:** Bonds of issuers in or expected to file for bankruptcy

Bid-ask spreads are narrower for liquid, on-the-run, high-credit-quality issues; wider for less liquid, seasoned, or distressed issues.

4 Fixed-Income Markets for Corporate Issuers

4.1 Short-Term Funding Alternatives

LOS 50.a: Compare short-term funding alternatives available to corporations and financial institutions.

For Nonfinancial Corporations:

Type	Description
Uncommitted Line of Credit	Bank may refuse to lend; flexible, no fees outside interest
Committed Line of Credit	Bank commits for specific period; commitment fee charged; renewal risk
Revolving Line of Credit	Longer term (years); restrictive covenants; most reliable
Secured (Asset-Backed) Loans	Backed by collateral (fixed assets, receivables, inventory)
Factoring	Transfer of credit granting and collection to lender at discount
Commercial Paper (CP)	Short-term unsecured debt; lower cost; rollover risk

For Financial Institutions:

- Deposits (demand, savings, time), Certificates of Deposit (CDs)
- Interbank funds, Central bank funds
- Commercial paper and Asset-Backed Commercial Paper (ABCP)

4.2 Repurchase Agreements (Repos)

LOS 50.b: Describe repurchase agreements (repos), their uses, and their benefits and risks.

A **repurchase agreement (repo)** is a collateralized loan where one party sells securities to another with a commitment to repurchase them later at a higher price.

$$\text{Purchase Price} = \text{Market Value of Securities}_{\overline{\text{Initial Margin}}}$$

$$\text{Repurchase Price} = \text{Purchase Price} \times \left[1 + \text{Repo Rate} \times \frac{\text{Days}}{360}\right]$$

$$\text{Haircut} = 1 - 1_{\overline{\text{Initial Margin}}}$$

Example: Repo Calculation: Bond $1M, margin 103%, rate 2%, 90 days

- Purchase price = 1,000,000 / 1.03 = $970,874
- Repurchase price = 970,874 × [1 + 0.02×90/360] = $975,728
- Haircut = 1 - 1/1.03 = 2.91%

Variation Margin: Additional collateral required if collateral value falls below required level.

Repo Applications: Finance positions, earn repo rate, monetary policy, short selling (reverse repo).

Repo Risks: Default risk, collateral risk, margining risk, legal risk, netting risk.

4.3 Investment-Grade vs High-Yield Issuers

LOS 50.c: Contrast the long-term funding of investment-grade versus high-yield corporate issuers.

Characteristic	**Investment-Grade**	**High-Yield**
Credit spread	Smaller proportion	Larger proportion
Primary concern	Ratings downgrade risk	Default risk
Covenants	Few; primarily affirmative	Many; negative covenants
Collateral	Usually unsecured	Often secured
Maturity flexibility	High; multiple maturities	Low; typically ≤10 years
Returns	Bond-like	More equity-like

5 Fixed-Income Markets for Government Issuers

5.1 Funding Choices for Government Issuers

LOS 51.a: Describe funding choices by sovereign and non-sovereign governments, quasi-government entities, and supranational agencies.

Sovereign Debt:

- Backed by taxing power; highest credit rating in domestic market
- **Developed market:** Stable economies, reserve currency, transparent fiscal policy
- **Emerging market:** Faster growth, less stable, commodity-dependent revenues
- **Domestic debt:** Issued in home currency to domestic investors
- **External debt:** Owed to foreign creditors; may be in foreign reserve currency

Non-Sovereign Issuers:

- **Agency/Quasi-government bonds:** Issued by government-created entities (e.g., Ginnie Mae)
- **General obligation (GO) bonds:** Backed by local taxing power
- **Revenue bonds:** Repaid from project-specific fees
- **Supranational bonds:** Issued by World Bank, IMF, etc.

5.2 Government vs Corporate Issuance and Trading

LOS 51.b: Contrast the issuance and trading of government and corporate fixed-income instruments.

Government Bond Auctions:

- Regular public auctions

- **Competitive bids:** Set the price (yield)
- **Noncompetitive bids:** Guaranteed allocation at auction price
- **Single-price auction:** All investors pay cut-off yield price
- **Multiple-price auction:** Successful bidders pay their bid price

Primary Dealers: Required to make competitive bids, submit bids for third parties, act as counterparties to central bank.

6 Fixed-Income Bond Valuation: Prices and Yields

6.1 Calculating Bond Price Given YTM

LOS 52.a: Calculate a bond's price given a yield-to-maturity on or between coupon dates.

$$\text{Price} = \text{C}\frac{}{1+YTM+\frac{C}{(1+YTM)^2}+\cdots+\frac{C+FV}{(1+YTM)^N}}$$

Example: 5-year, 10% annual coupon, YTM=10%: N=5, PMT=10, FV=100, I/Y=10, CPT PV = 100 Example: Same bond with YTM=8%: CPT PV = 107.99 (premium) Example: Same bond with YTM=12%: CPT PV = 92.79 (discount)

Semiannual Coupon Bond:

$$\text{Price} = \sum_{t=1}^{2N} \frac{C/2}{(1+YTM/2)^t} + \frac{FV}{(1+YTM/2)^{2N}}$$

Accrued Interest:

$$\text{AI} = \text{Coupon} \times \frac{\text{Days since last coupon}}{\text{Days in coupon period}}$$

Flat Price vs Full Price:

- **Flat price (clean price):** Quoted price without accrued interest
- **Full price (dirty price):** Flat price + accrued interest

6.2 Bond Price Relationships

LOS 52.b: Identify the relationships among a bond's price, coupon rate, maturity, and yield-to-maturity.

Key Relationships:

1. **Inverse price-yield relationship:** Yield ↑ → Price ↓; Yield ↓ → Price ↑
2. **Lower coupon → greater price sensitivity** to yield changes
3. **Longer maturity → greater price sensitivity** to yield changes (usually)
4. **Convex relationship:** Price increase from yield decrease >Price decrease from equal yield increase
5. **Pull to par:** Price converges to par as maturity approaches

Example: Premium bond: Coupon >YTM → Price >Par Example: Discount bond: Coupon <YTM → Price <Par Example: Par bond: Coupon = YTM → Price = Par

6.3 Matrix Pricing

LOS 52.c: Describe matrix pricing.

Matrix pricing is a method for estimating YTM (or price) of bonds that are not traded or infrequently traded.

Example: 6-year bond, 7% coupon. Comparables: 5-year (6.40%), 8-year (7.20%)

- Interpolated YTM = 6.40% + (1/3)(7.20%-6.40%) = 6.67%
- 7% coupon ¿ 6.67% YTM → Bond trades at premium

7 Yield and Yield Spread Measures for Fixed-Rate Bonds

7.1 Annual Yield for Varying Compounding Periods

LOS 53.a: Calculate annual yield on a bond for varying compounding periods in a year.

$$\text{Effective Annual Yield} = \left(1 + \frac{YTM}{n}\right)^n - 1$$

Example: YTM=10%, semiannual (n=2): (1+0.10/2)^2 - 1 = 10.25% Example: YTM=10%, quarterly (n=4): (1+0.10/4)^4 - 1 = 10.38%

Key Yield Measures:

Measure	Formula/Description
Current Yield	Annual coupon / Price
Simple Yield	(Annual coupon + Annual straight-line amortization) / Price
Yield to Call (YTC)	YTM using call date and call price
Yield to Worst (YTW)	Lowest of YTM and all yields to call

7.2 Yield Spread Measures

LOS 53.b: Compare, calculate, and interpret yield and yield spread measures for fixed-rate bonds.

Benchmark Spreads:

- **G-spread:** Yield spread over government bond yield
- **I-spread:** Yield spread over swap rate
- **Z-spread (Zero-Volatility Spread):** Constant spread added to each spot rate to make PV = market price
- **OAS (Option-Adjusted Spread):** Spread after removing embedded option value

OAS = Z-spread - Option Value (in bp)

For callable bonds, OAS <Z-spread (option value subtracted). For putable bonds, OAS >Z-spread.

8 Yield and Yield Spread Measures for Floating-Rate Instruments

8.1 Floating-Rate Note Yields

LOS 54.a: Calculate and interpret yield spread measures for floating-rate instruments.

FRN Terminology:

- **Quoted Margin (QM):** Fixed margin above MRR actually paid
- **Discount Margin (DM):** Margin required to price FRN at par
- QM = DM → FRN trades at par
- DM >QM → FRN trades at discount
- DM <QM → FRN trades at premium

Example: FRN Valuation: $100k, semiannual, MRR=3%, QM=120bp, DM=150bp, 5 years

- Coupon rate = 3% + 1.2% = 4.2% → semiannual payment = 2.1%
- Required return = 3% + 1.5% = 4.5% → semiannual discount rate = 2.25%
- N=10, I/Y=2.25, PMT=2.1, FV=100 → PV = 98.67 (discount)

8.2 Money Market Instrument Yields

LOS 54.b: Calculate and interpret yield measures for money market instruments.

Yield Type	**Formula**
Add-on Yield (365-day)	HPY × (365 / Days)
Discount Yield (360-day)	(Discount / Face) × (360 / Days)
Money Market Yield (MMY)	(Discount / Price) × (360 / Days)
Bond Equivalent Yield (BEY)	HPY × (365 / Days)

Example: 90-day T-bill, discount yield = 1.2%:

- Discount = 1.2% × 90/360 × 1,000 = \$3 → Price = \$997
- HPY = 3/997 = 0.3009%
- BEY = 0.3009% × 365/90 = 1.2203%

9 The Term Structure of Interest Rates

9.1 Spot Rates and Spot Curve

LOS 55.a: Define spot rates and the spot curve, and calculate the price of a bond using spot rates.

Spot rates are discount rates for single payments to be received in the future (zero-coupon rates).

$$\text{Bond Price} = \text{C}\frac{}{(1+S_1)+\frac{C}{(1+S_2)^2}+\cdots+\frac{C+FV}{(1+S_N)^N}}$$

Example: 3-year, 5% annual coupon bond, S1=3%, S2=4%, S3=5%: Price = $5/1.03 + 5/1.04^2 + 105/1.05^3 = 100.18$

9.2 Par Rates and Forward Rates

LOS 55.b: Define par and forward rates, and calculate par rates, forward rates from spot rates, spot rates from forward rates, and bond price using forward rates.

Par rate is the coupon rate that would make a bond trade at par given the spot curve.

Forward rate is an interest rate for a loan to be made at a future date. Notation: 1y1y = 1-year loan one year from now.

$$\boxed{(1+\mathrm{S}_2)^2 = (1+S_1)(1+1y1y)}$$

$$\boxed{1\mathrm{y}1\mathrm{y} = (1+\mathrm{S}_2)^2 \frac{}{1+S_1-1}}$$

Example: S1=4%, S2=8%: 1y1y = (1.08²/1.04) - 1 = 12.154%

$$\boxed{\mathrm{S}_3 = [(1+S_1)(1+1y1y)(1+2y1y)]^{1/3} - 1}$$

9.3 Comparing Spot, Par, and Forward Curves

LOS 55.c: Compare the spot curve, par curve, and forward curve.

Normal/Upward-Sloping Curve: Forward rates >Spot rates >Par yields

Inverted/Downward-Sloping Curve: Forward rates <Spot rates <Par yields

Flat Curve: All rates equal across maturities.

10 Interest Rate Risk and Return

10.1 Sources of Return

LOS 56.a: Calculate and interpret the sources of return from investing in a fixed-rate bond.

Three Sources of Return:

1. Coupon and principal payments
2. Reinvestment income on coupons
3. Capital gain/loss if sold before maturity

Key Results (assuming reinvestment rate = YTM):

1. Hold to maturity with unchanged YTM → Return = YTM
2. Sell before maturity with unchanged YTM → Return = YTM
3. YTM increases, hold to maturity → Return >original YTM
4. YTM decreases, hold to maturity → Return <original YTM
5. Short horizon: Price risk dominates; Long horizon: Reinvestment risk dominates

10.2 Macaulay Duration

LOS 56.b: Describe the relationships among a bond's holding period return, its Macaulay duration, and the investment horizon.

$$\boxed{\text{Duration Gap} = \text{Macaulay Duration - Investment Horizon}}$$

- Positive gap → Exposed to price risk from increasing rates
- Negative gap → Exposed to reinvestment risk from decreasing rates
- Zero gap → Price and reinvestment risk offset

LOS 56.c: Define, calculate, and interpret Macaulay duration.

Macaulay duration is the weighted average of time until each promised cash flow, using PV of cash flows as weights.

$$\boxed{\text{MacDur} = \sum_{t=1}^{N} t \times PV(CF_t)_{\overline{\text{Price}}}}$$

Example: 5-year, 11% annual coupon, YTM=15%, Price=86.59: MacDur = 4.03 years

11 Yield-Based Bond Duration Measures and Properties

11.1 Modified Duration, Money Duration, PVBP

LOS 57.a: Define, calculate, and interpret modified duration, money duration, and PVBP.

$$\boxed{\text{ModDur} = \text{MacDur}_{\overline{1+YTM/n}}}$$

$$\boxed{\text{Approx ModDur} = \text{V}_{-} - V_{+\overline{2V_0\Delta YTM}}}$$

$$\boxed{\%\Delta P \approx -\text{ModDur} \times \Delta YTM}$$

$$\boxed{\text{Money Duration} = \text{ModDur} \times \text{Full Price}}$$

$$\boxed{\text{PVBP} = \text{V}_{-} - V_{+\overline{2\text{ (per \$100 par) or Money Duration}\times 0.0001}}}$$

11.2 Factors Affecting Interest Rate Risk

LOS 57.b: Explain how a bond's maturity, coupon, and yield level affect its interest rate risk.

Effects on Duration:

- Longer maturity → Higher duration (usually)
- Higher coupon → Lower duration
- Higher YTM → Lower duration
- Passage of time → Duration decreases smoothly, jumps up slightly on coupon dates

12 Yield-Based Bond Convexity and Portfolio Properties

12.1 Convexity and Convexity Adjustment

LOS 58.a: Calculate and interpret convexity and describe the convexity adjustment.

$$\text{Convexity (single cash flow)} = \text{t(t+1)}\frac{}{(1+r)^2}$$

$$\text{Approx Convexity} = \text{V}_- + V_+ - 2V_0\frac{}{(\Delta YTM)^2 V_0}$$

12.2 Percentage Price Change with Duration and Convexity

LOS 58.b: Calculate the percentage price change of a bond for a specified change in yield, given the bond's duration and convexity.

$$\%\Delta P \approx -\text{ModDur} \times \Delta YTM + \frac{1}{2} \times \text{Convexity} \times (\Delta YTM)^2$$

Example: ModDur=3.50, Convexity=16.9, YTM = -0.5%:

- Duration effect = -3.50 × -0.005 = 1.75%
- Convexity effect = 0.5 × 16.9 × $(0.005)^2$ = 0.0211%
- Total = 1.7711% → New price = 86.59 × 1.017711 = 88.125

LOS 58.c: Calculate portfolio duration and convexity and explain the limitations.

$$\text{Portfolio Duration} = \sum w_i \times D_i$$

$$\text{Portfolio Convexity} = \sum w_i \times C_i$$

Portfolio duration assumes parallel shifts in the yield curve (all yields change by same amount).

13 Curve-Based and Empirical Fixed-Income Risk Measures

13.1 Effective Duration and Convexity

LOS 59.a: Explain why effective duration and effective convexity are the most appropriate measures for bonds with embedded options.

$$\boxed{\text{EffDur} = \text{V}_- - V_+ \overline{2V_0 \Delta\text{Curve}}}$$

$$\boxed{\text{EffCon} = \text{V}_- + V_+ - 2V_0 \overline{(\Delta\text{Curve})^2 V_0}}$$

Bonds with embedded options have uncertain cash flows and no single well-defined YTM. Effective measures use benchmark curve shifts.

LOS 59.b: Calculate percentage price change using effective duration and convexity.

$$\boxed{\%\Delta P \approx -\text{EffDur} \times \Delta\text{Curve} + \tfrac{1}{2} \times \text{EffCon} \times (\Delta\text{Curve})^2}$$

Callable bonds and MBS may exhibit negative convexity at low yields (price increase slows as yields fall).

13.2 Key Rate Duration

LOS 59.c: Define key rate duration and describe its use to measure price sensitivity to benchmark yield curve changes.

Key rate duration measures the sensitivity of a bond/portfolio to changes in benchmark yield for a specific maturity, holding other yields constant.

Uses: Measure shaping risk (nonparallel yield curve shifts). Sum of key rate durations = effective duration.

13.3 Empirical vs Analytical Duration

LOS 59.d: Describe the difference between empirical duration and analytical duration.

Analytical Duration: Calculated mathematically from bond characteristics.

Empirical Duration: Estimated from historical relationship between benchmark yield changes and bond price changes.

Empirical duration may be lower than analytical duration during "flight to quality" when government yields fall but corporate spreads widen.

14 Credit Risk

14.1 Components of Credit Risk

LOS 60.a: Describe credit risk and its components, probability of default and loss given default.

Credit risk is the risk of losses from a borrower failing to service debt (make interest/principal payments).

$$\boxed{\text{Expected Loss} = \text{Probability of Default} \times \text{Loss Given Default}}$$

$$\boxed{\text{LGD\%} = 1 - \text{Recovery Rate}}$$

$$\boxed{\text{Credit Spread} \approx \text{POD} \times \text{LGD\%}}$$

Example: POD=3%, Recovery=75%: Spread = 0.03 × 0.25 = 0.75%. Actual spread = 1.5% → More than fairly compensated.

14.2 Credit Ratings

LOS 60.b: Describe the uses of ratings from credit rating agencies and their limitations.

Moody's	S&P/Fitch	Category
Aaa	AAA	Highest quality
Aa1-Aa3	AA+ to AA-	High quality
A1-A3	A+ to A-	Upper medium
Baa1-Baa3	BBB+ to BBB-	Investment grade
Ba1-Ba3	BB+ to BB-	Non-investment grade

B1-B3	B+ to B-	Highly speculative
Caa-C	CCC-C	Substantial risk

Limitations: Lag market pricing; some risks hard to assess; split ratings possible; mistakes occur.

14.3 Factors Influencing Yield Spreads

LOS 60.c: Describe macroeconomic, market, and issuer-specific factors that influence yield spreads.

Macroeconomic: Economic cycle affects spreads; high-yield spreads more volatile.

Market (Liquidity Risk): Wider bid-ask spreads → higher liquidity risk. Liquidity spread = bid yield - offer yield.
Example: Decomposing Yield Spread: Bid=99.5, Offer=100.5, Coupon=5%, Benchmark=3% Midprice yield = 5% → total spread = 2% Bid yield = 5.065%, Offer yield = 4.935% → liquidity spread = 0.13% Credit spread = 2% - 0.13% = 1.87%

15 Credit Analysis for Government Issuers

LOS 61.a: Explain special considerations when evaluating the credit of sovereign and non-sovereign government debt issuers.

Sovereign Credit Analysis - Qualitative:

1. Institutions and policy (political stability, legal protections)
2. Fiscal flexibility (ability to increase taxes or decrease spending)
3. Monetary effectiveness (central bank independence)
4. Economic flexibility (growth trends, income per capita, diversity)
5. External status (reserve currency, geopolitical risks)

Sovereign Credit Analysis - Quantitative:

1. Fiscal strength (debt/GDP, debt/revenue, interest/revenue)

2. Economic growth and stability (real GDP growth, per-capita GDP)
3. External stability (foreign exchange reserves/GDP, reserves/external debt)

Non-Sovereign Issuers: GO bonds (backed by taxing power) vs Revenue bonds (backed by project fees)

16 Credit Analysis for Corporate Issuers

16.1 Qualitative and Quantitative Factors

LOS 62.a: Describe the qualitative and quantitative factors used to evaluate a corporate borrower's creditworthiness.

Qualitative: Business model, industry competition, business risk, corporate governance, covenants, accounting policies.

Quantitative: Strong profits, low leverage, high coverage ratios, high liquidity → higher credit quality.

16.2 Financial Ratios for Credit Analysis

LOS 62.b: Calculate and interpret financial ratios used in credit analysis.

Ratio	Formula
EBIT Margin	EBIT / Revenue
EBIT/Interest	EBIT / Interest expense
Debt/EBITDA	Total debt / EBITDA
Debt/Capital	Total debt / (Total debt + Equity)
RCF/Net Debt	Retained cash flow / (Total debt - Cash)

Example: York: EBIT margin 25% >Zale 20.5%; York Debt/EBITDA 2.5× >Zale 0.9× → York lower credit quality

16.3 Seniority Rankings

LOS 62.c: Describe the seniority rankings of debt, secured vs unsecured debt, and priority of claims in bankruptcy.

Priority of Claims (Highest to Lowest):

1. First lien/mortgage
2. Senior secured (second lien)
3. Junior secured
4. Senior unsecured
5. Senior subordinated
6. Subordinated
7. Junior subordinated

Notching: Adjusting issue credit rating up/down from issuer rating based on seniority, collateral, structural subordination.

17 Fixed-Income Securitization

17.1 Benefits of Securitization

LOS 63.a: Explain benefits of securitization for issuers, investors, economies, and financial markets.

Securitization Process:

1. Originator creates pool of debt-based assets
2. Pool sold to Special Purpose Entity (SPE)
3. SPE issues Asset-Backed Securities (ABS) to investors

Benefits to Originators: Increased lending, improved profitability, lower capital reserves, improved liquidity.

Benefits to Investors: Tailored risk/return, access to collateral, liquidity.

Benefits to Markets: Decreased liquidity risk, improved efficiency, lower financing costs.

17.2 Parties in Securitization

LOS 63.b: Describe securitization, including the parties and the roles they play.

Party	**Role**
Seller/Depositor	Originates assets, sells to SPE
Issuer/Trust (SPE)	Buys assets, issues ABS
Servicer	Collects payments, administers loans
Trustee	Oversees safekeeping of collateral

SPE is bankruptcy remote; originator's financial problems don't affect ABS investors.

18 Asset-Backed Security (ABS) Instrument and Market Features

18.1 Covered Bonds

LOS 64.a: Describe characteristics and risks of covered bonds and how they differ from other asset-backed securities.

Covered Bonds: Senior debt with dual recourse (cover pool + issuer's unencumbered assets), dynamic pool, lower yields than ABS.

18.2 Credit Enhancement Structures

LOS 64.b: Describe typical credit enhancement structures used in securitizations.

Internal Credit Enhancements:

- Overcollateralization: Collateral value >ABS face value
- Excess spread: Income from collateral >coupon to investors
- Credit tranching (subordination): Multiple classes with different priority

Example: Credit Tranching:

- Tranche A (senior): $300M, lowest yield, highest rating
- Tranche B (subordinated): $80M, medium yield
- Tranche C (equity/junior): $30M, highest yield, first to absorb losses

18.3 Non-Mortgage ABS Types

LOS 64.c: Describe types and characteristics of non-mortgage asset-backed securities.

Credit Card ABS: Nonamortizing; lockout/revolving period (interest only); amortization period (principal passed through).

Solar ABS: Backed by solar installation loans; attractive to ESG investors; pre-funding period.

18.4 Collateralized Debt Obligations (CDOs)

LOS 64.d: Describe collateralized debt obligations, including their cash flows and risks.

CDO Types:

- CBO (Collateralized Bond Obligation): Corporate/emerging market debt
- CLO (Collateralized Loan Obligation): Leveraged bank loans (most common)
- Cash flow CLO: Payments from collateral cash flows
- Market value CLO: Payments from trading collateral
- Synthetic CLO: Credit derivative contracts

19 Mortgage-Backed Security (MBS) Instrument and Market Features

19.1 Prepayment Risk

LOS 65.a: Define prepayment risk and describe time tranching structures in securitizations.

Prepayment risk is uncertainty about prepayment speeds of underlying mortgages.

Components:

- **Contraction risk:** Prepayments faster than expected (rates fall)
- **Extension risk:** Prepayments slower than expected (rates rise)

Time Tranching (Sequential Pay): Principal flows to tranches in order; short tranche has more contraction risk; long tranche has more extension risk.

19.2 Residential Mortgage Loan Features

LOS 65.b: Describe fundamental features of residential mortgage loans that are securitized.

Key Features:

- Prepayment penalties (common in Europe, rare in US)
- Recourse vs nonrecourse loans
- **LTV (Loan-to-Value):** Lower LTV → less risk
- **DTI (Debt-to-Income):** Lower DTI → less risk

Agency vs Non-Agency RMBS:

- Agency: Guaranteed by government/GSE; meet underwriting standards
- Non-agency: Private entities; include credit enhancements

19.3 Residential MBS Types

LOS 65.c: Describe types and characteristics of residential mortgage-backed securities.

Mortgage Pass-Through Security:

- **WAM (Weighted Average Maturity):** Weighted average of remaining maturities
- **WAC (Weighted Average Coupon):** Weighted average of mortgage rates

Example: WAM = 210×(90/409) + 100×(72/409) + 280×(247/409) = 233 months Example: WAC = 2.6%×(90/409) + 1.0%×(72/409) + 5.4%×(247/409) = 4.0%

CMO Tranche Types: Sequential pay, PAC, support, Z-tranche, PO, IO, floating-rate, residual.

19.4 Commercial MBS

LOS 65.d: Describe characteristics and risks of commercial mortgage-backed securities.

CMBS Features:

- Backed by income-producing real estate
- Fewer mortgages in pool (less diversification)
- Analysis focuses on property credit risk

$$\boxed{\text{DSCR} = \text{Net Operating Income}_{\overline{\text{Debt Service}}}}$$

$$\boxed{\text{LTV} = \text{Mortgage Amount}_{\overline{\text{Appraised Value}}}}$$

Call Protection: Loan-level (lockout, penalty points, defeasance) and structure-level (sequential pay).

Balloon Risk: Partially amortizing; balloon payment at maturity creates extension risk if borrower can't refinance.

Part II

Derivatives

20 Derivative Instrument and Derivative Market Features

20.1 Derivative Definition and Features

LOS 66.a: Define a derivative and describe basic features of a derivative instrument.

A derivative is a security whose value derives from the value of another security or variable (the underlying) at a specific future date.

Basic Features: Underlying, forward price, settlement date, contract size, zero value at initiation (for forwards/futures/swaps).

Hedging vs Speculating: Hedge transfers/offsets existing risk; Speculate increases risk.

20.2 Derivative Markets

LOS 66.b: Describe basic features of derivative markets, and contrast OTC and exchange-traded derivative markets.

Exchange-Traded: Standardized, central clearinghouse (CCH), daily mark-to-market, margin deposits, liquid, transparent, lower counterparty risk.

OTC Derivatives: Customized, no central location, less regulated, counterparty credit risk (unless central clearing mandate applies).

21 Forward Commitment and Contingent Claim Features

21.1 Forward, Futures, Swaps, Options, CDS

LOS 67.a: Define forward contracts, futures contracts, swaps, options, and credit derivatives and compare their basic characteristics.

Forward: Agreement to buy/sell asset at specific price on future date.

Futures: Exchange-traded, standardized, daily mark-to-market, margin deposits, central clearinghouse.

Swap: Exchange series of payments; interest rate swap: fixed vs floating.

Options: Call (right to buy), Put (right to sell). Buyer pays premium.

Credit Default Swap (CDS): Protection buyer pays fixed; protection seller pays if credit event occurs.

21.2 Option Value at Expiration and Profit

LOS 67.b: Determine the value at expiration and profit from a long or a short position in a call or put option.

$$\boxed{\mathrm{c}_T = \max(0, S_T - X)}$$

$$\boxed{\mathrm{p}_T = \max(0, X - S_T)}$$

$$\boxed{\pi_{\text{call}} = \max(0, S_T - X) - c_0}$$

$$\boxed{\pi_{\text{put}} = \max(0, X - S_T) - p_0}$$

Example: Call: X=40, premium=3, $S_T = 35ßc = 0, profit = -3$

21.2 Forward Commitment vs Contingent Claim

LOS 67.c: Contrast forward commitments with contingent claims.

Forward Commitment: *Legally binding promise (forwards, futures, most swaps).*

Contingent Claim: *Payoff depends on event (options, CDS).*

22 Derivative Benefits, Risks, and Uses

22.1 Benefits and Risks

LOS 68.a: Describe benefits and risks of derivative instruments.

Benefits: *Change/transfer risk, information discovery, operational advantages (easy short sales, low costs, leverage, liquidity), improved market efficiency.*

Risks: *Implicit leverage, basis risk, liquidity risk, counterparty credit risk, systemic risk.*

22.2 Uses by Issuers and Investors

LOS 68.b: Compare the use of derivatives among issuers and investors.

Issuer Uses (Hedge Accounting):

- *Cash flow hedge: Hedges future cash flow variability*
- *Fair value hedge: Hedges changes in asset/liability values*
- *Net investment hedge: Hedges foreign subsidiary equity value*

Investor Uses: *Modify risk exposure, hedge, speculate, create synthetic positions.*

23 Arbitrage, Replication, and Cost of Carry

23.1 Arbitrage and Replication

LOS 69.a: Explain how the concepts of arbitrage and replication are used in pricing derivatives.

$$\boxed{F_0(T) = S_0(1 + R_f)^T}$$

Example: Arbitrage: If $F > S(1+Rf)^T : Sell forward, buy underlying, borrow at Rf. If F$
$Rf)^T : Buy forward, short sell, lend at Rf.$

23.2 Spot vs Expected Future Price and Cost of Carry

LOS 69.b: Explain the difference between spot and expected future price and the cost of carry.

$$\boxed{F_0(T) = [S_0 + PV(costs) - PV(benefits)](1 + R_f)^T}$$

$$\boxed{F_0(T) = S_0 e^{(r+c-b)T}}$$

Forward Exchange Rate: $F = S \times \frac{1+R_{price}}{1+R_{base}}$

24 Pricing and Valuation of Forward Contracts

LOS 70.a: Explain how the value and price of a forward contract are determined at initiation, during the life, and at expiration.

At Initiation: $F_0(T) = S_0(1 + R_f)^T$, *Value = 0*
During Life: $V_t(T) = S_t - F_0(T)(1 + R_f)^{-(T-t)}$
At Expiration: $V_T(T) = S_T - F_0(T)$

24.1 Forward Rate Agreements (FRAs)

LOS 70.b: Explain how forward rates are determined for interest rate forward contracts and describe their uses.

$$\boxed{FRA\ Payoff = Notional \times \frac{Reference\ Rate - Forward\ Rate}{1 + Reference\ Rate \times Period} \times Period}$$

25 Pricing and Valuation of Futures Contracts

LOS 71.a: Compare the value and price of forward and futures contracts.

Differences: *Forward: constant price, fluctuating value, settlement at expiration. Futures: daily mark-to-market resets value to zero each day.*

LOS 71.b: Explain why forward and futures prices differ.

Pricing: *If interest rates constant or uncorrelated → prices equal. Positive correlation → futures more attractive. Negative correlation → forwards more attractive.*

26 Pricing and Valuation of Interest Rate Swaps

LOS 72.a: Describe how swap contracts are similar to but different from a series of forward contracts.

Similarity: *Swap can be replicated by series of FRAs with contract rate = swap fixed rate.*

Difference: *Individual FRAs may not have zero value; sum of FRA values = zero at initiation.*

LOS 72.b: Contrast the value and price of swaps.

Price: *Fixed rate (par swap rate)*

Value: $V = PV(\text{expected floating}) - PV(\text{fixed})$, *zero at initiation, changes with expectations.*

27 Pricing and Valuation of Options

27.1 Exercise Value, Moneyness, Time Value

LOS 73.a: Explain the exercise value, moneyness, and time value of an option.

Moneyness: *In-the-money: exercise generates positive payoff; Out-of-the-money: loss; At-the-money: zero.*

$$\boxed{\textit{Call exercise value} = \max(0, S - X)}$$

$$\boxed{\textit{Put exercise value} = \max(0, X - S)}$$

$$\boxed{\textit{Option premium} = \textit{Exercise value} + \textit{Time value}}$$

27.2 Arbitrage and Replication for Options

LOS 73.b: Contrast the use of arbitrage and replication concepts in pricing forward commitments and contingent claims.

Upper Bounds: $c_0 \leq S_0$, $p_0 \leq X(1 + R_f)^{-T}$

Lower Bounds: $c_0 \geq \max[0, S_0 - X(1 + R_f)^{-T}]$, $p_0 \geq \max[0, X(1 + R_f)^{-T} - S_0]$

27.3 Factors Affecting Option Value

LOS 73.c: Identify the factors that determine option value and describe how each affects value.

Factor ↑	***Call Value***	***Put Value***
Underlying price (S)	↑	↓
Exercise price (X)	↓	↑
Risk-free rate	↑	↓
Volatility	↑	↑
Time to expiration	↑ *(usually)*	↑ *(usually)*
Benefits (dividends)	↓	↑
Storage costs	↑	↓

28 Option Replication Using Put-Call Parity

LOS 74.a: Explain put-call parity for European options.

$$\boxed{c_0 + X(1+R_f)^{-T} = S_0 + p_0}$$

Fiduciary call: $c + X(1+R_f)^{-T}$ ***Protective put:*** $S + p$
Synthetic Securities:

- *Synthetic stock:* $S_0 = c + X(1+R_f)^{-T} - p$
- *Synthetic call:* $c = S + p - X(1+R_f)^{-T}$
- *Synthetic put:* $p = c + X(1+R_f)^{-T} - S$

LOS 74.b: Explain put-call forward parity for European options.

$$\boxed{F_0(T)(1+R_f)^{-T} + p_0 = c_0 + X(1+R_f)^{-T}}$$

29 Valuing a Derivative Using a One-Period Binomial Model

LOS 75.a: Explain how to value a derivative using a one-period binomial model.

$$h = c^+ - c^- \frac{}{S^+ - S^-}$$

Option Value Process:

1. *Find hedge ratio h*
2. *Set $hS^+ - c^+ = hS^- - c^-$, solve for h*
3. *Risk-free portfolio value $= hS - c$*
4. *Discount at risk-free rate: $V_0 = (hS - c) = (hS^+ - c^+)/(1 + R_f)$*
5. *Solve for c*

LOS 75.b: Describe the concept of risk neutrality in derivatives pricing.

$$\pi = \frac{1+R_f-d}{u-d}$$

$$c_0 = \frac{\pi c^+ + (1-\pi)c^-}{1+R_f}$$

Risk-neutral probabilities are derived from arbitrage relationships, not actual probabilities.

Appendix: Complete Formula Summary

Fixed Income Formulas

Bond Price: $P = \sum \frac{C}{(1+YTM)^t} + \frac{FV}{(1+YTM)^N}$
Accrued Interest: $AI = Coupon \times \frac{Days\ since\ last\ coupon}{Days\ in\ coupon\ period}$
Effective Annual Yield: $EAY = (1 + YTM/n)^n - 1$
Forward Rate: $1y1y = \frac{(1+S_2)^2}{1+S_1} - 1$
Modified Duration: $ModDur = \frac{MacDur}{1+YTM/n}$
Convexity Adjustment: $\%\Delta P = -ModDur \times \Delta YTM + \frac{1}{2}C \times (\Delta YTM)^2$
Money Duration: $MoneyDur = ModDur \times P$
PVBP: $PVBP = MoneyDur \times 0.0001$
Expected Loss: $EL = POD \times (1 - Recovery\ Rate)$
Repo Purchase Price: $PP = \frac{Market\ Value}{Initial\ Margin}$

Derivatives Formulas

Forward Price (no costs): $F_0(T) = S_0(1 + R_f)^T$
Forward Price (with costs/benefits): $F_0(T) = [S_0 + PV(costs) - PV(benefits)](1 + R_f)^T$
Forward Value: $V_t(T) = S_t - F_0(T)(1 + R_f)^{-(T-t)}$
Call Payoff: $c_T = \max(0, S_T - X)$
Put Payoff: $p_T = \max(0, X - S_T)$
Put-Call Parity: $c + X(1 + R_f)^{-T} = S + p$
Hedge Ratio: $h = \frac{c^+ - c^-}{S^+ - S^-}$
Risk-Neutral Probability: $\pi = \frac{1+R_f-d}{u-d}$
Binomial Option Value: $c_0 = \frac{\pi c^+ + (1-\pi)c^-}{1+R_f}$

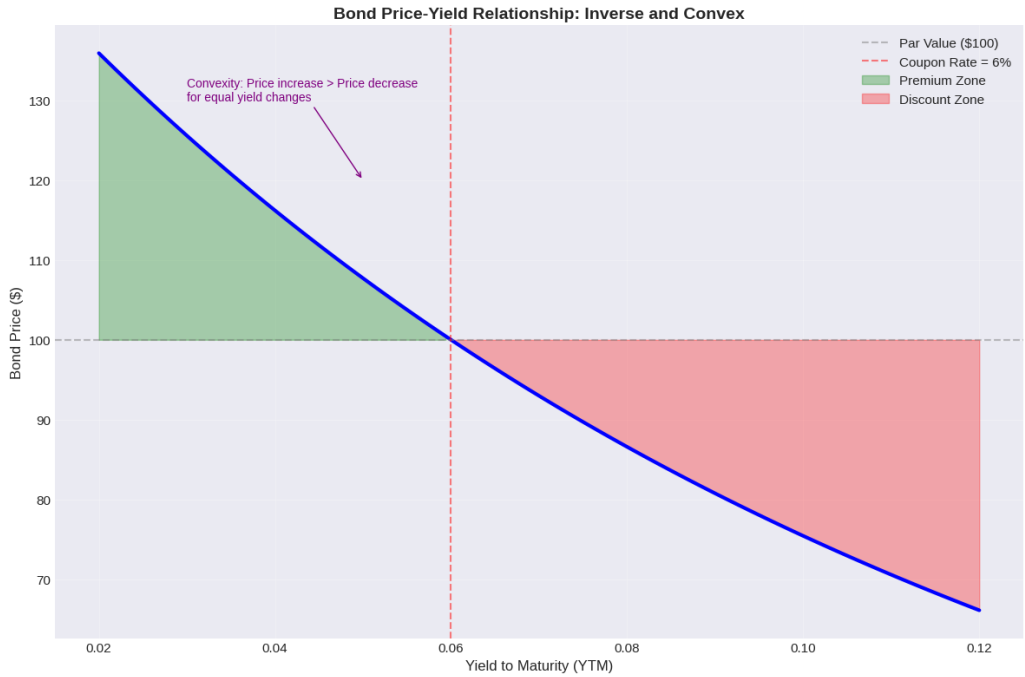

Figure 1: Bond Price-Yield Relationship: Inverse and Convex - When yields rise, bond prices fall (and vice versa), with convexity causing larger price increases than decreases for equal yield changes

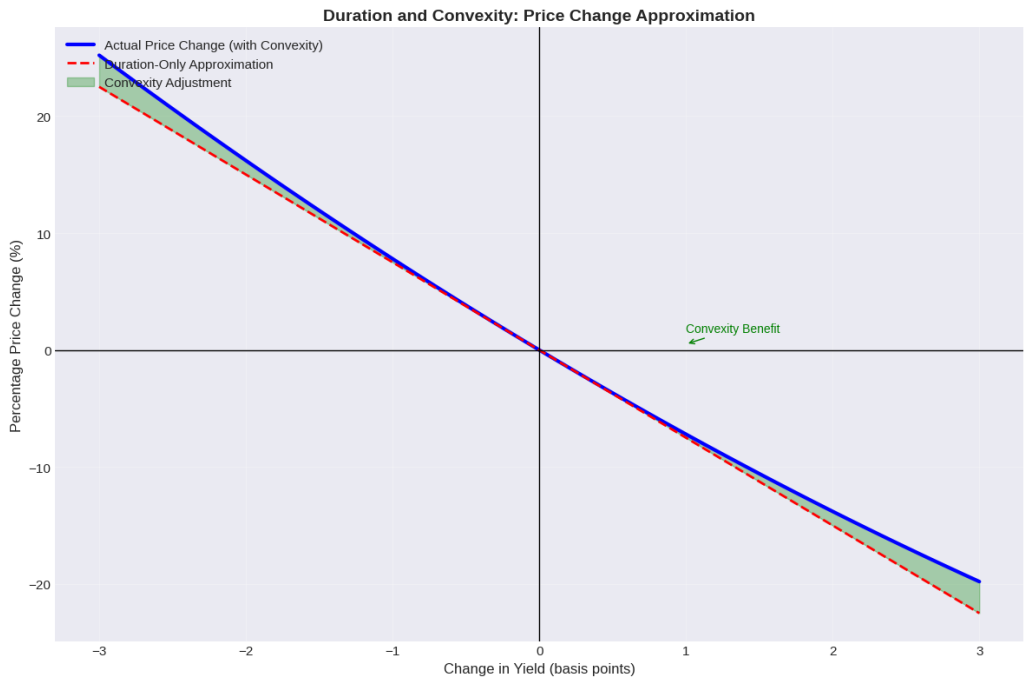

Figure 2: Duration and Convexity: Price Change Approximation - Duration provides a linear estimate, while convexity captures the curvature for more accurate price changes

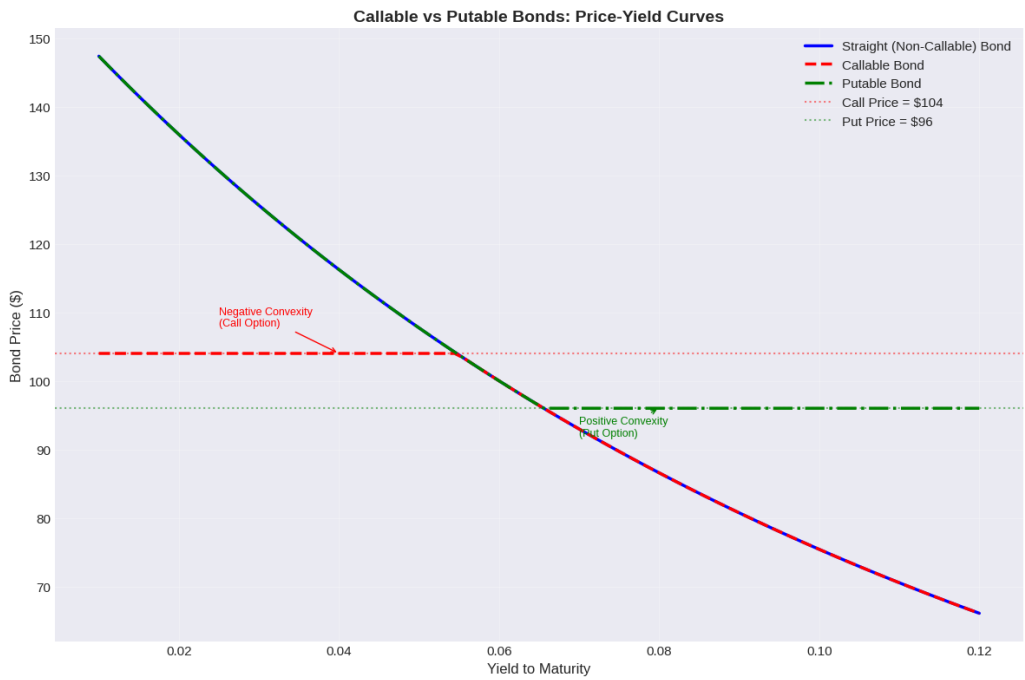

Figure 3: Callable vs Putable Bonds: Price-Yield Curves - Callable bonds exhibit negative convexity (price compression), while putable bonds show positive convexity (price support)

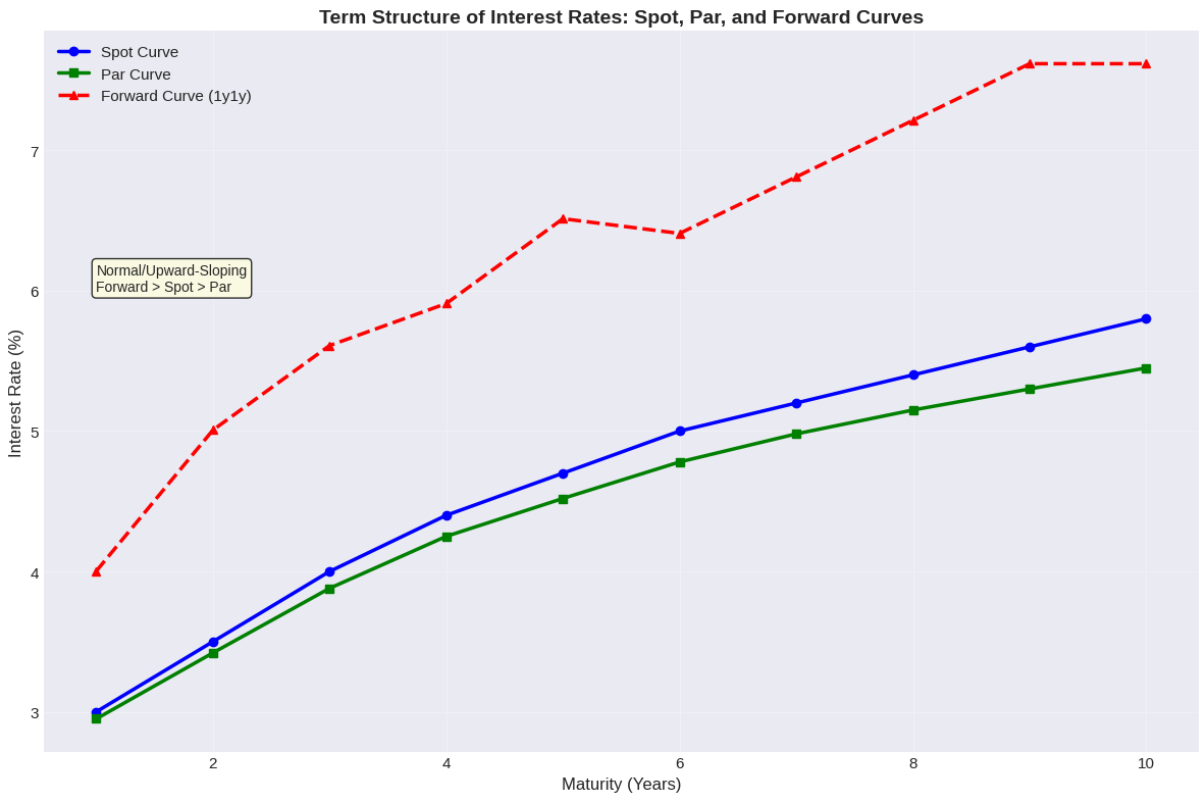

Figure 4: Term Structure of Interest Rates: Spot, Par, and Forward Curves - In a normal upward-sloping environment, forward rates ¿ spot rates ¿ par yields

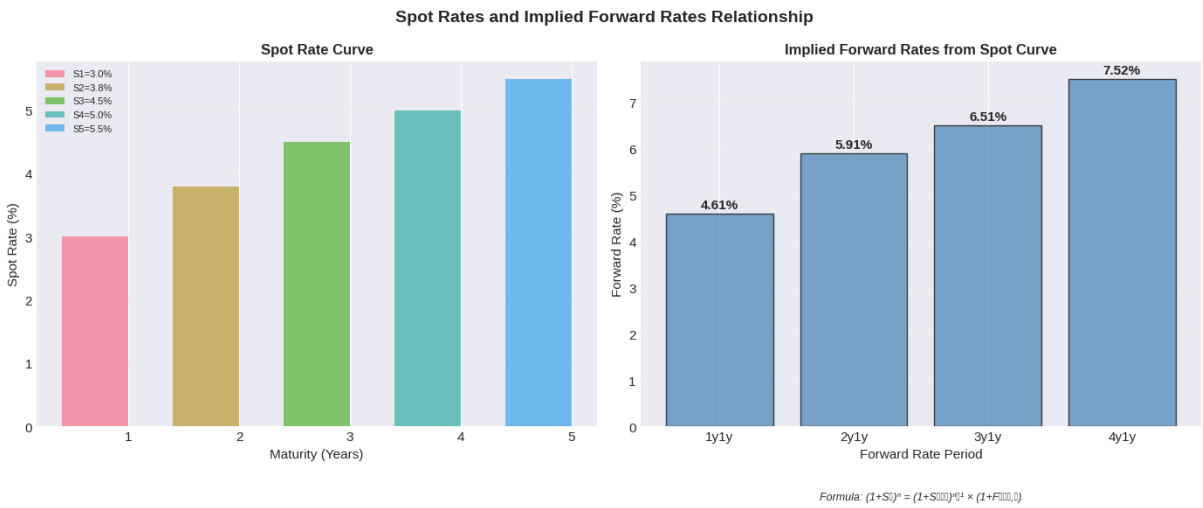

Figure 5: Spot Rates and Implied Forward Rates Relationship - Forward rates are derived mathematically from the spot curve using arbitrage-free pricing

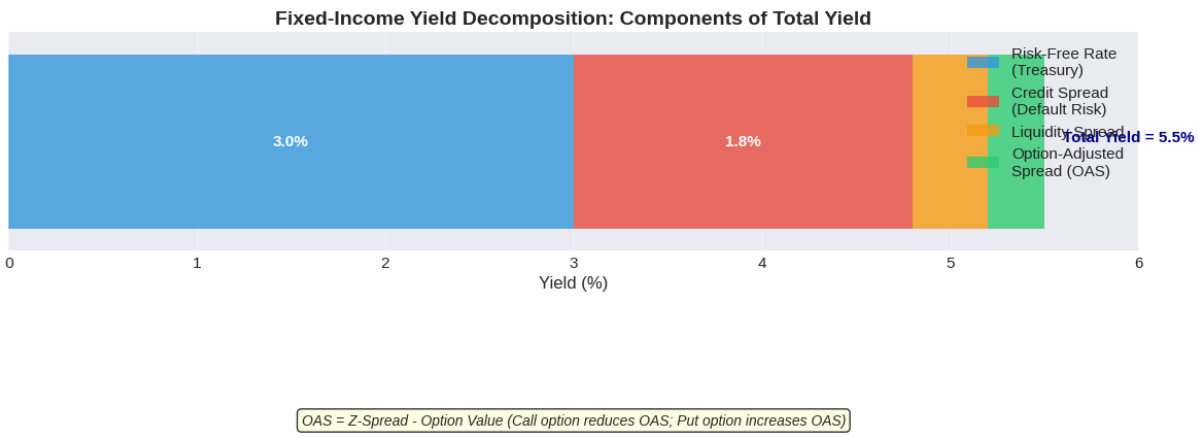

Figure 6: Fixed-Income Yield Decomposition: Components of Total Yield - Total yield comprises risk-free rate, credit spread, liquidity spread, and option-adjusted spread

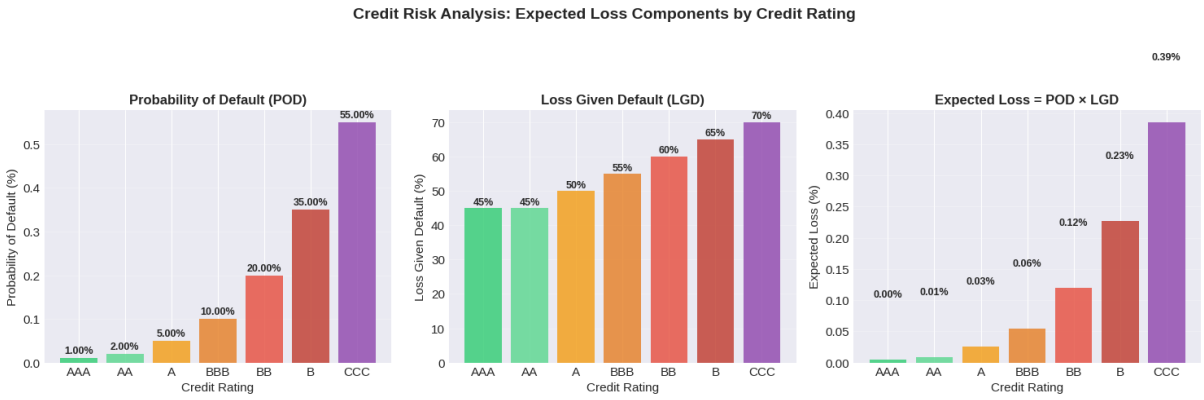

Figure 7: Credit Risk Analysis: Expected Loss Components by Credit Rating - Expected Loss = Probability of Default (POD) × Loss Given Default (LGD)

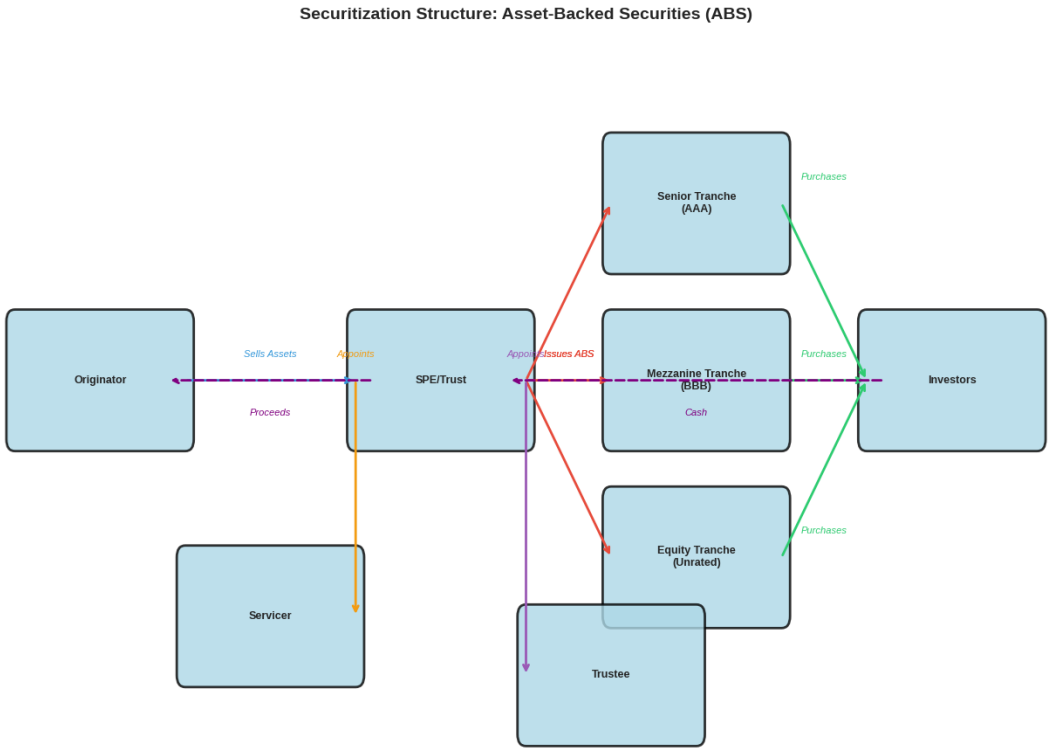

Figure 8: Securitization Structure: Asset-Backed Securities (ABS) - Special Purpose Entity (SPE) purchases assets from originator and issues tranched securities to investors

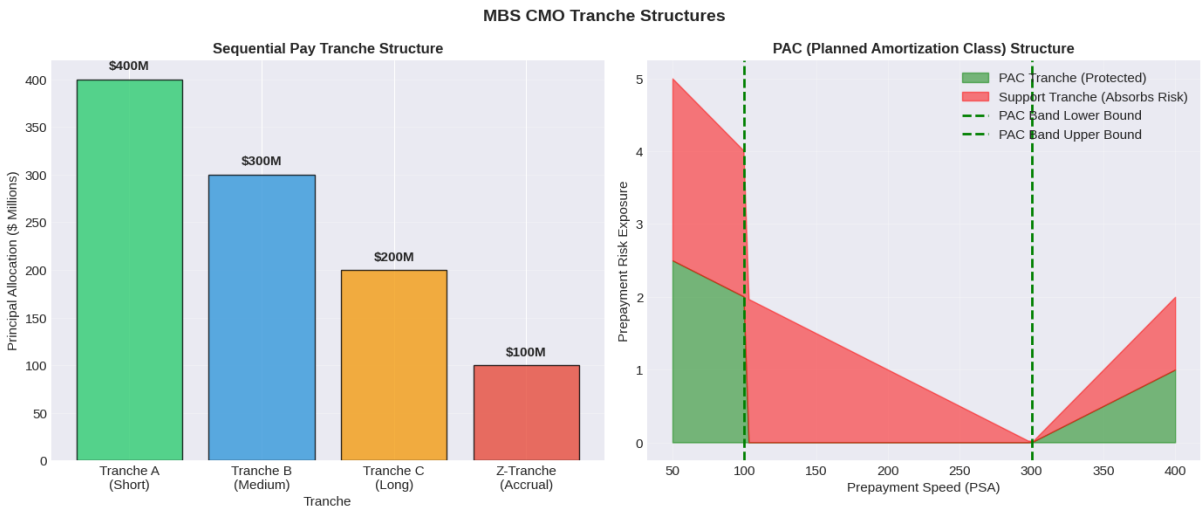

Figure 9: MBS CMO Tranche Structures - Sequential pay distributes principal in order; PAC tranches provide prepayment protection within specified bands

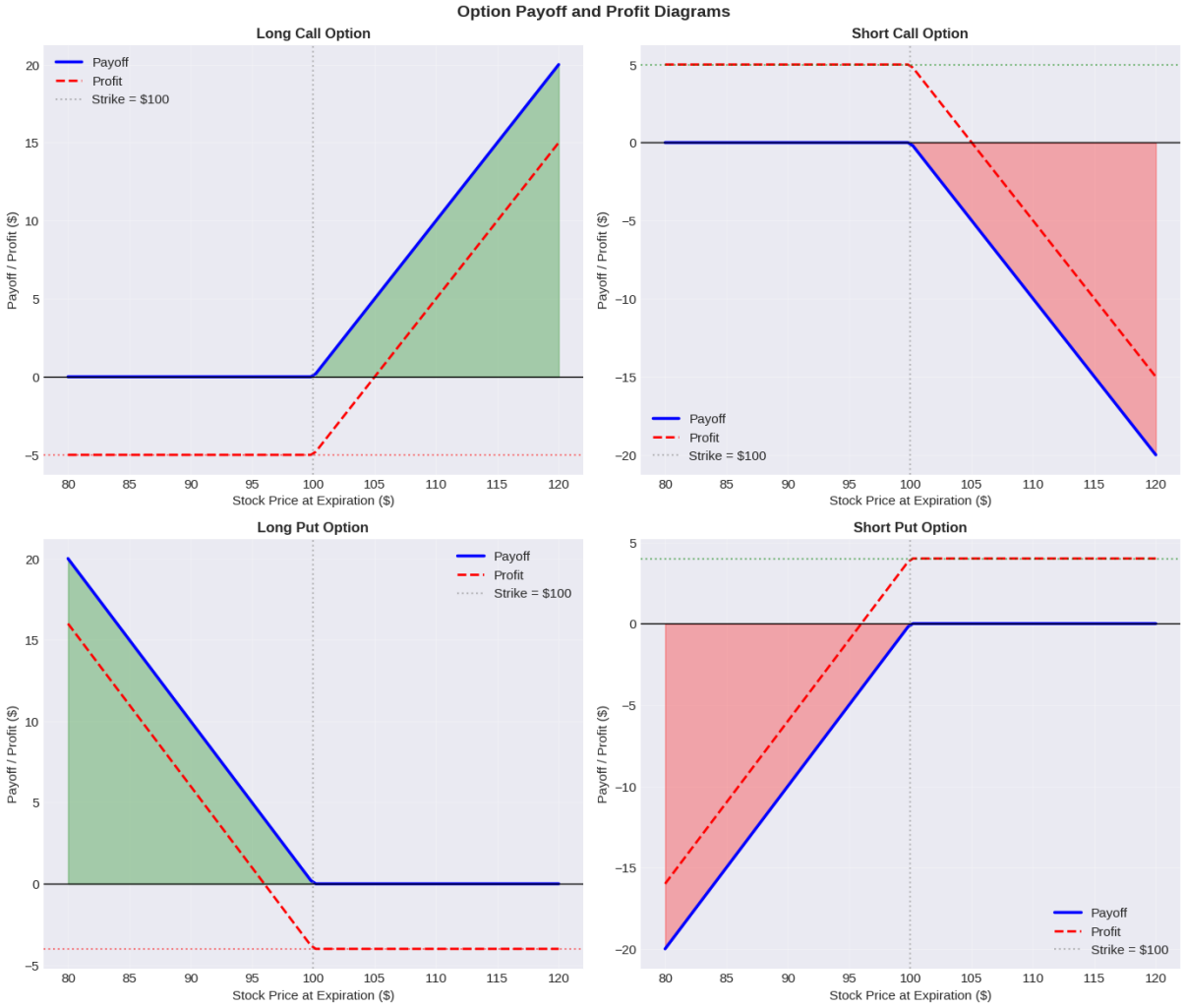

Figure 10: Option Payoff and Profit Diagrams - Long call (unlimited upside), short call (limited upside, unlimited downside), long put (limited downside), short put (limited upside, significant downside)

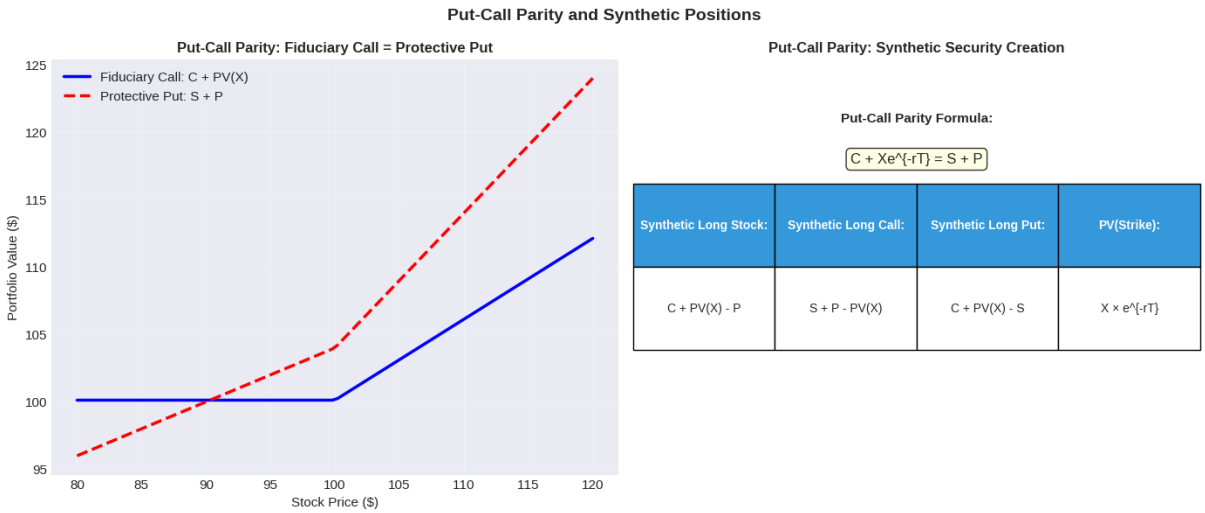

Figure 11: Put-Call Parity and Synthetic Positions - Fiduciary Call (C + PV(X)) equals Protective Put (S + P), enabling synthetic security creation

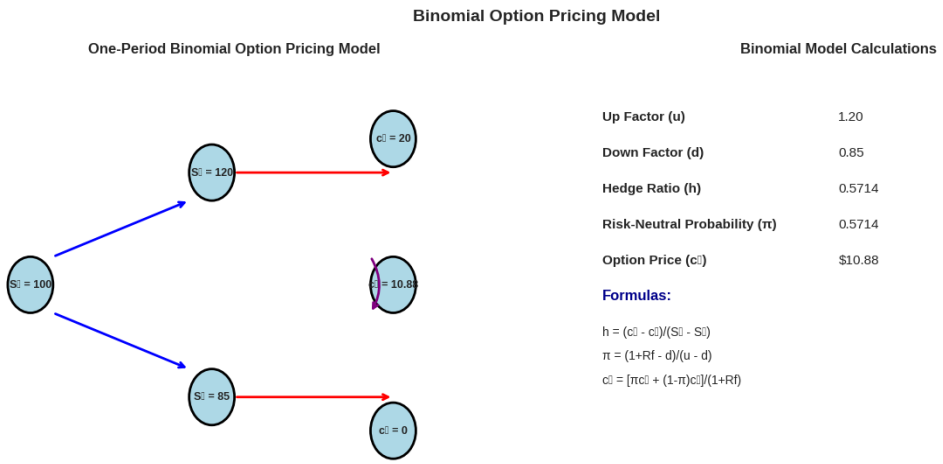

Figure 12: One-Period Binomial Option Pricing Model - Uses risk-neutral probability and hedge ratio to value options in a simplified two-state framework

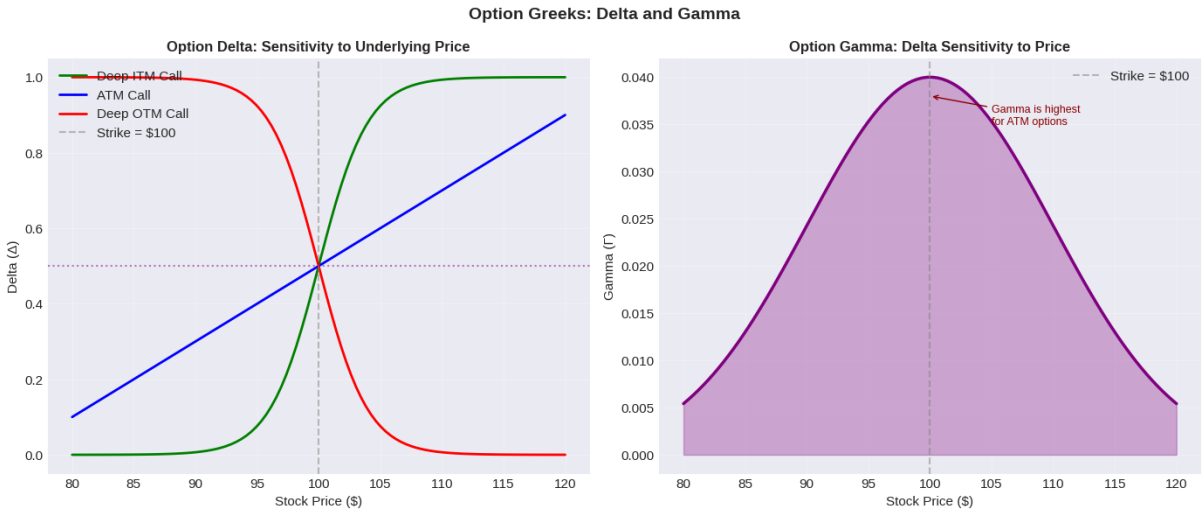

Figure 13: Option Greeks: Delta and Gamma - Delta measures price sensitivity to underlying; Gamma measures Delta's sensitivity (highest for at-the-money options)

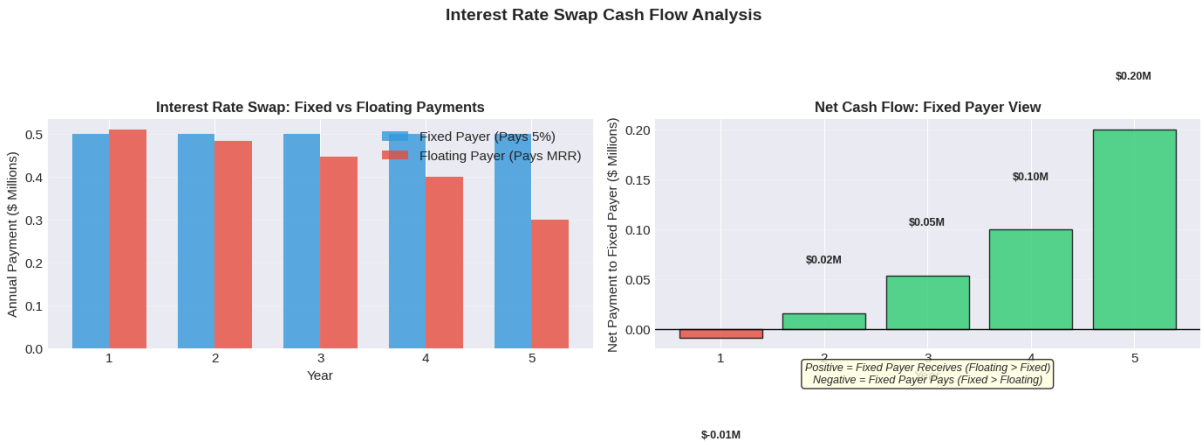

Figure 14: Interest Rate Swap Cash Flow Analysis - Fixed payer pays fixed rate and receives floating; net cash flow depends on whether floating exceeds fixed